

VARDO, Norway

WMO: 010980

Lat: 70.371N

Lon: 31.096E

Elev: 15

StdP: 101.14

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-12.8	-10.8	-14.5	1.1	-11.8	-12.9	1.2	-9.8	19.9	-4.4	18.3	-4.1	9.4	210	0.883

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	3.6	16.8	13.9	15.0	12.6	13.5	11.5	14.1	16.4	12.8	14.6	11.8	13.2	4.4	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.8	9.2	15.3	11.7	8.6	13.9	10.8	8.1	12.7	39.7	16.6	36.3	14.8	33.7	13.2	20.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.3	14.3	12.7	DB	-14.7	20.3	2.5	2.4	-16.5	22.0	-17.9	23.4	-19.3	24.7	-21.1	26.4	(4)
(5)			WB	-14.8	16.4	2.4	1.7	-16.6	17.6	-18.0	18.6	-19.4	19.6	-21.1	20.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	2.4	-3.7	-4.3	-2.5	0.2	3.5	6.9	10.0	10.0	7.8	3.4	-0.4	-2.2	(6)	
	DBStd	5.79	3.39	3.57	2.76	2.79	2.72	2.46	2.56	2.29	2.34	2.81	2.80	3.16	(7)	
	HDD10.0	2838	426	401	388	293	203	99	32	28	73	204	312	377	(8)	
	HDD18.3	5807	684	634	647	543	461	343	259	259	316	463	562	636	(9)	
	CDD10.0	72	0	0	0	0	1	6	31	28	7	0	0	0	(10)	
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(11)	
	CDH23.3	1	0	0	0	0	0	0	1	0	0	0	0	0	(12)	
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)	
(14) Wind	WSAvg	6.2	8.1	8.1	7.5	5.9	4.9	4.6	4.2	4.1	5.2	6.3	7.2	7.9	(14)	
(15) Precipitation	PrecAvg	578	54	45	41	35	33	43	50	54	49	64	55	55	(15)	
	PrecMax	831	167	123	125	90	76	109	130	142	114	143	159	124	(16)	
	PrecMin	315	17	7	5	9	9	5	5	14	14	17	15	4	(17)	
	PrecStd	101	25	24	21	17	17	23	32	26	19	25	27	22	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.2	4.1	4.6	8.1	13.4	16.9	20.3	19.8	15.0	10.9	6.8	5.3	(19)	
		MCWB	2.9	2.6	2.7	5.6	9.0	13.5	15.5	15.9	11.9	9.3	5.5	3.5	(20)	
	2%	DB	2.9	2.7	3.2	5.8	10.5	13.7	17.3	16.4	13.3	9.4	5.0	3.8	(21)	
		MCWB	1.9	1.4	1.7	4.0	7.6	11.1	14.3	13.6	11.3	7.9	3.9	2.6	(22)	
	5%	DB	1.8	1.7	2.2	4.6	8.8	11.7	15.5	14.5	12.1	8.3	4.1	2.9	(23)	
		MCWB	0.7	0.5	0.9	3.1	6.5	9.5	13.1	12.3	10.4	7.1	3.0	1.7	(24)	
	10%	DB	0.7	0.6	1.4	3.7	7.3	10.2	13.7	13.2	11.1	7.3	3.2	2.0	(25)	
		MCWB	-0.3	-0.5	0.2	2.5	5.4	8.3	11.9	11.4	9.7	6.2	2.1	1.0	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.1	2.9	3.0	5.6	9.5	13.6	16.2	16.2	12.7	9.5	5.7	3.8	(27)	
		MCDB	3.8	3.9	4.2	7.9	12.5	16.3	19.3	18.5	14.0	10.6	6.8	4.8	(28)	
	2%	WB	2.0	1.6	1.9	4.2	7.8	11.4	14.6	14.0	11.6	8.1	4.1	2.7	(29)	
		MCDB	2.8	2.5	3.0	5.5	10.2	13.3	17.0	16.1	13.0	9.2	4.8	3.6	(30)	
	5%	WB	0.9	0.6	1.1	3.3	6.6	9.7	13.3	12.6	10.6	7.2	3.2	1.9	(31)	
		MCDB	1.6	1.5	2.1	4.3	8.5	11.4	15.3	14.2	11.8	8.2	4.0	2.8	(32)	
	10%	WB	-0.2	-0.4	0.3	2.7	5.6	8.5	12.0	11.7	9.8	6.3	2.3	1.1	(33)	
		MCDB	0.6	0.5	1.3	3.6	7.1	9.9	13.5	12.9	10.9	7.2	3.1	2.0	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	4.0	3.9	3.3	3.2	3.4	3.4	3.7	3.6	3.4	2.9	3.3	3.6	(35)	
		MCDBR	4.6	4.1	3.7	3.7	6.1	6.5	7.3	6.2	4.9	3.4	3.3	3.9	(36)	
	5% WB	MCWBR	4.3	3.7	3.3	2.8	4.1	4.4	4.9	4.3	3.5	3.0	3.2	3.5	(37)	
		MCWBR	4.5	4.1	3.6	3.4	5.8	6.3	7.0	5.7	4.3	3.2	3.1	3.8	(38)	
(40) Clear-Sky Solar Irradiance	taub		N/A	0.265	0.315	0.337	0.352	0.344	0.360	0.346	0.315	0.270	N/A	N/A	(40)	
		taud	N/A	2.087	2.213	2.301	2.383	2.484	2.471	2.491	2.512	2.177	N/A	N/A	(41)	
	Ebn at Noon		0	425	672	790	827	850	817	785	705	412	0	0	(42)	
		Edh at Noon	0	38	72	91	95	89	88	76	56	33	0	0	(43)	
(44) All-Sky Solar Radiation	RadAvg	0.00	0.24	1.34	2.98	4.16	4.74	4.31	2.91	1.53	0.41	0.00	0.00	0.00	(44)	
	RadStd	0.00	0.03	0.10	0.23	0.46	0.33	0.39	0.29	0.14	0.04	0.00	0.00	0.00	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	+0.67	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)	
(47) Regional (10 neighbors)	+0.43	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)	

Nomenclature: See separate page